

Snapshot Week 01 of Group PG1

Multi-Ammunition Soft-Kill System (MASS) Project

		Team Members	
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Course: Software Development

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1 Product Backlog and Task Board

1.1 Product Backlog

Placeholder: Trello screenshot of product backlog, including tasks: “Implement I/O system,” “Add LHS sampling,” “Build linear regression model,” “Test output format,” “Document architecture,” and future tasks like “Add Bayesian Optimization.” Prioritized for Sprint 1.

Figure 1: Product Backlog

1.2 Task Board

Placeholder: Trello screenshot showing Sprint 1 task board with columns: To Do (“Test output format”), In Progress (“Implement I/O system”), Done (“Set up Git repo”).

Figure 2: Task Board

2 Sprint Backlog and User Stories

2.1 Sprint Backlog

Placeholder: Trello screenshot of Sprint 1 backlog: “Implement I/O system (3 points),” “Add LHS sampling (5 points),” “Build linear regression model (4 points),” “Test output format (2 points),” “Document architecture (2 points).”

Figure 3: Sprint Backlog

2.2 User Stories

The following user stories, selected for Sprint 1, focus on building a prototype for the MASS competition:

- As a Navy officer, I want intelligent sampling to learn from limited queries so that simulations are efficient. Tasks: Implement LHS sampling, plan for Bayesian Optimization.
- As a competition judge, I want predictions as space-separated floats with minimal squared error to score submissions accurately. Tasks: Build linear regression model, test output format.
- As a developer, I want a robust stdout/stdin query system to interact reliably with the simulator. Tasks: Implement I/O system, test with mock data.
- As a team member, I want a prioritized backlog to focus Sprint 1 on core features. Tasks: Document backlog, assign tasks.

3 Definition of Done

Our Definition of Done ensures Sprint 1 deliverables meet quality standards:

- Code is committed to GitHub Enterprise, reviewed, and tested.
- Prototype handles 1D/2D inputs, queries the simulator, and outputs space-separated predictions.
- Unit tests pass for sampling and output formatting.
- Documentation (architecture, research summary) is complete in the Git repo.
- Sprint deliverables (code, docs, presentation) are prepared for review.

4 Summary of Changes

This first snapshot for Group PG1 covers Week 1 of Sprint 1 (August 14–28, 2025). We set up a GitHub Enterprise repository ([Insert URL]) with `src/` (code), `docs/` (Agile artifacts), and `tests/` (unit tests). A prototype in `src/main.py` parses command-line inputs (dimensions, array size, max queries), uses Latin Hypercube Sampling (LHS), and outputs predictions via linear regression. We created `docs/user_stories.md`, `backlog.md`, and

[1] “Dynamic soft-kill weapon-target assignment,” ResearchGate, 2024.